ENGO 625 – Advanced GNSS Theory and Applications (F2025)

Lab #5 – Carrier Phase Ambiguity Resolution
(Instructor: Kyle O'Keefe)

Due Date

Monday, 15 December 2025 at 4:30 PM. Please submit your written report and source code.

Objectives

1. To become familiar with phase processing and ambiguity resolution.
2. To understand the abilities of carrier phase differential processing using a fixed solution in L1 only mode.
3. To improve programming skills with regards to Geomatics Engineering (C/C++, MATLAB or Python, please).

Data Description

Use the data you were provided from Lab 1-4.

Tasks

1. Modify your code from Lab 4 to resolve the L1 ambiguities after the solution has converged. Justify your criteria for sufficient convergence. Use a sum-of-squares search method (nothing fancy, a brute force search over the search volume is sufficient) where you keep track of the ambiguity set with the smallest and second smallest weighted sum-of-squares of change in ambiguity.
   a. List the estimated integer values of the ambiguities and the corresponding ratio test value.
   b. For each ambiguity, list the search space used for the ambiguity search.
   c. Plot the fixed ambiguities on top of the float ambiguities (from Lab 4) at each epoch. Centre the accuracy envelope on the (changing) float ambiguity value. Comment on the behavior of the float ambiguities over time, and on the agreement between the float and fixed ambiguities.
2. Use the L1 fixed integer ambiguities to determine the position of the rover receiver. Then:
   a. Repeat Task 3 of Lab 4 (ie plot the position errors and their uncertainty).
   b. Comment on the change in accuracy that fixed ambiguities provide compared to float ambiguities.
3. Choose one of the ambiguities at random, and add exactly **one cycle** to its fixed value. Use this incorrect fixed-integer ambiguity and the other correct fixed-integer ambiguities to determine the position of the rover receiver.
   a. Repeat subtasks 2a, 2b and 2c of Lab 4 but this time with the one wrong ambiguity.
   b. Plot the phase residuals (if using SLSQ) or innovations (if using KF) over time.
   c. Comment on the effect of a single incorrect cycle on the accuracy and how the phase residuals/innovations can be used to detect such an occurrence.

Report Format

Each report must be professionally written. Be concise but thorough. Explanations should accompany all plots. Assumptions must be clearly documented and justified.

Grading Criteria (7% of Final Grade)

Technical Accuracy (5/7)

- Correct solution/plots
- Correct analysis of results
- Correct justification and assumptions
- Insight into results
- Graphs correctly labeled (each axis with out a unit or title will be -1 point)

Source Code (1/7)

- Methodical and logical procession of algorithms
- Commented source code
- If you share software with other classmates, you must acknowledge their contribution both in the comments and in the text of the report.

Report Quality, Neatness and Readability (1/7)

- Neat, clear and effective use scales on plots
- Logical organization of lab

References

ENGO 625 Course Lecture Notes - Advanced GNSS Theory and Applications
ENGO 421 Course Lecture Notes (might be useful for coordinate systems)
ENGO 363 Course Lecture Notes or Text (for a detailed explanation of Least Squares)